

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461106.8603 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1528 +/- 0.0077
Transit depth (Rp/Rs)^2	2.33 +/- 0.23 [%]
Orbital Inclination (inc)	89.67 +/- 1.57
Airmass coefficient 1 (a1)	1.459 +/- 0.014
Airmass coefficient 2 (a2)	-0.36 +/- 0.1
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#3 - [344, 81]
Optimal Aperture	3.83
Optimal Annulus	15.81
Transit Duration (day)	0.0859 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1528 $\pm$ 0.0077 vs 0.126 (deviation 3.12 σ)
Duration fit vs published	0.0859 d vs 0.0925 d (deviation 2.04 σ)
Mid-time O-C	7.7 min
Depth SNR	17.8
Residual scatter	0.94 %

Light curve

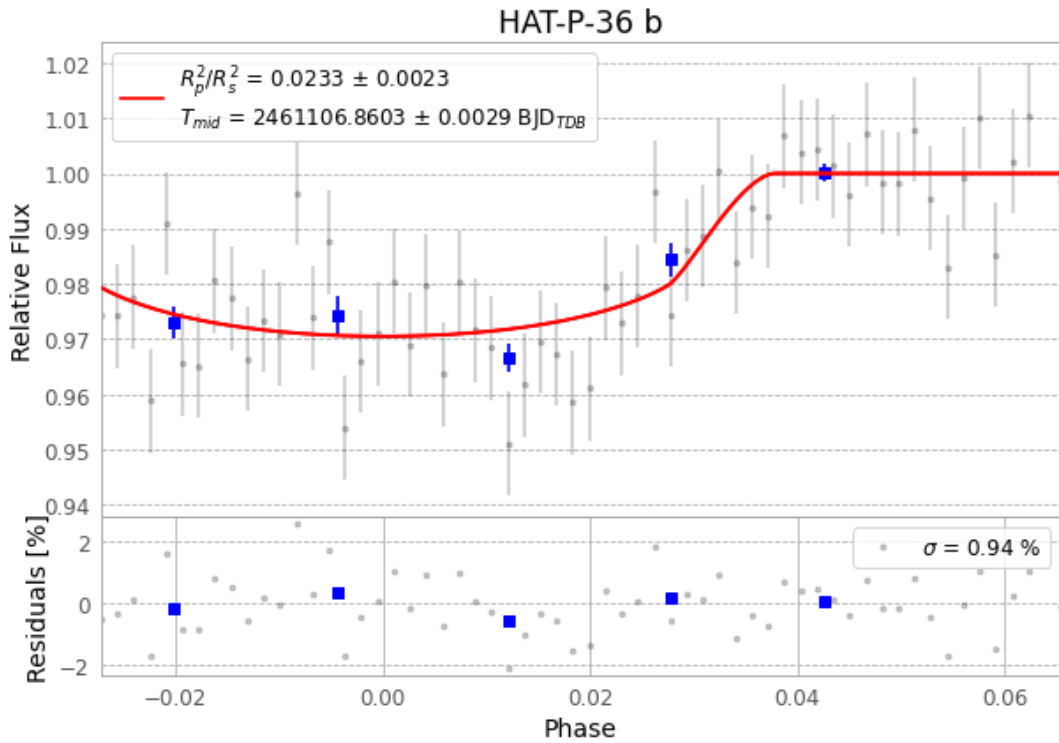


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-03-07.png (SHA-256 20b5e4af41bbfc2e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 97], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2752aa0928fb47b6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

59 FITS frames (39 MB), HATP-36260307073916.FITS → HATP-36260307103615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260307.log (edfe4464bd84...), FinalLightCurve_HAT-P-36 b_2026-03-07.png (20b5e4af41bb...), FinalLightCurve_HAT-P-36 b_2026-03-07.csv (75b352de3b44...)

Compute resources

Wall time 175.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 01:20Z.